Science — Unit 3 — Study Guide Practice · Grade 6

The Repair Café Challenge

Answer 15 short questions in about 10–15 minutes. Choose one answer or write the missing word. Check your answers at the end.

> You help at a neighbourhood repair café. Follow materials from the workshop to the garden and help the team make less waste.

1 Which list contains only raw materials found in nature?

- Wood, oil and sand
- A chair, a bottle and a window
- A notebook, a can and a jacket
- Paper, a spoon and a toy

> **Answer:** Wood, oil and sand
> Wood, oil and sand are natural resources used to make products.

2 What happens when waste is recycled?

- It is buried and left in a landfill
- Its materials are processed to make new products
- It is stored without being changed
- It is always burned to produce heat

> **Answer:** Its materials are processed to make new products
> Recycling recovers materials from waste and turns them into new products.

3 In the garden, tiny living things break down organic waste. The finished material added to the soil is called _____.

> **Answer:** compost
> Composting transforms organic waste into compost, which improves soil.

4 Which plan sorts these materials correctly? The local collection program accepts the empty plastic container.

- Recycle the food scraps; compost the tin can and plastic container
- Compost all the materials together
- Recycle the tin can and plastic container; compost the apple core and healthy seed-free plants
- Recycle the plants; throw away the tin can and plastic container

> **Answer:** Recycle the tin can and plastic container; compost the apple core and healthy seed-free plants
> Fruit scraps and suitable garden plants are compostable. Tin cans and accepted plastic containers can be recycled through the appropriate collection program.

5 A mill collects old cardboard. Name the thin material used for writing that it can make from the recovered fibres.

> **Answer:** paper or recycled paper
> Fibres recovered from cardboard can be used to make new paper.

6 A factory recovers plastic from used PET bottles. Which product can it make for winter clothing?

- Cotton fabric
- Wool yarn
- Leather
- Polar fleece

> **Answer:** Polar fleece
> Recovered PET plastic can become polyester fibres used in polar fleece.

7 Food scraps are sent to a well-managed composting site instead of a landfill. Why can this help the climate?

- It keeps all gases inside the food forever
- It can reduce methane released from landfilled food
- It turns plastic into clean air
- It stops every source of air pollution

> **Answer:** It can reduce methane released from landfilled food
> Food can produce methane as it breaks down without oxygen in a landfill. Properly managed composting uses oxygen and can reduce these emissions.

8 A farmer uses the material made at a composting site in a field. How can this reduce the need for chemical fertilizers?

- It removes all minerals from the soil
- It prevents every plant disease
- It provides some nutrients that plants need
- It replaces all the water needed by crops

> **Answer:** It provides some nutrients that plants need
> The material adds organic matter and nutrients. It can reduce the need for added fertilizers, but does not replace every crop need.

9 A manufacturer uses recovered materials instead of only new resources. What can this help save?

- Raw materials
- Landfill space only
- The need to sort waste
- Every kind of fuel completely

> **Answer:** Raw materials
> Recycling can reduce the need to extract new natural resources.

10 What benefit do recycling and composting share?

- They make every discarded object disappear
- They turn all waste into food for people
- They make sorting unnecessary
- They keep useful materials out of the garbage sent to landfills

> **Answer:** They keep useful materials out of the garbage sent to landfills
> Both recover useful materials and reduce the amount of garbage sent for disposal. Waste still needs to be prevented and sorted.

11 You clean an old mug and use it as a pencil holder. Which of the three Rs describes this action?

> **Answer:** reuse or reusing
> Reuse means using an object again without processing its material into a new product.

12 At a sorting centre, which object can an ordinary magnet pick up?

- An aluminium drink can
- A glass jar
- An ordinary steel food can
- A plastic cup

> **Answer:** An ordinary steel food can
> Ordinary steel food cans contain iron and are attracted to magnets. Aluminium, glass and plastic are not picked up this way.

13 In moist soil, which usually breaks down sooner, an uncoated paper bag or an ordinary plastic bag?

- The plastic bag
- The paper bag
- They always take the same time
- Neither can ever change

> **Answer:** The paper bag
> Paper can be broken down by living things. Ordinary plastic usually persists much longer. The time depends on the material and conditions.

14 Trucks take mixed recyclable containers to a facility where workers and machines sort them. Is this a landfill or a recycling plant?

> **Answer:** recycling plant or a recycling plant or recycling centre or a recycling centre or recycling center or a recycling center
> Sorting happens at a recycling plant or sorting centre. A landfill is used for waste disposal.

15 You cut a healthy basil stem and place its base in water to grow roots. What method are you using?

- Growing from seeds
- Pollination
- Composting
- Propagation with cuttings

> **Answer:** Propagation with cuttings
> A stem cutting can develop roots and grow into a plant. This is a form of asexual plant propagation.